Chirality of the Seeley–DeWitt coefficients and quartic Weyl structure in the spectral action

David Alfyorov

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Abstract

We prove that the traced Seeley–DeWitt coefficients $\mathrm{tr}(a_{2n})$ for the Dirac operator on a Ricci-flat four-manifold decompose as $\mathrm{tr}(a_{2n}) = f_n(p) + f_n(q)$, where $p = |C^+|^2$ and $q = |C^-|^2$ are the squared norms of the self-dual and anti-self-dual Weyl tensors. The proof rests on a *chirality theorem*: the spin connection generators $\sigma^{rs} = \frac{1}{4}[\gamma^r, \gamma^s]$ commute with the chirality operator γ_5 in four dimensions, rendering the curvature endomorphism $\Omega_{\mu\nu}$ block-diagonal in the chiral basis. We establish a *crossed chirality assignment*—the left-handed spinor block couples exclusively to the anti-self-dual Weyl tensor, and the right-handed block to the self-dual part—using the 't Hooft symbol decomposition. At the quartic level (mass dimension 8), the theorem implies that $\mathrm{tr}(a_8)$ contains only the combination $c(p^2 + q^2)$ with zero pq cross-term, fixing the ratio $(C^2)^2 : (\overset{*}{C}C)^2 = 1{:}1$ exactly. The result extends to the full Standard Model spectral triple on the almost-commutative geometry $M \times F$. By a Molien series analysis, the three parity-even quartic Weyl invariants reduce first to two via the Cayley–Hamilton identity, and then to one effective structure via chirality. This reduces the three-loop finiteness problem for the spectral action from an overdetermined system to a single ratio condition. We show that the strict spectral renormalizability conjecture for gravity is *disproven*: the graviton propagator is chirality-blind (a scalar on the traceless-transverse sector), and multi-eigenvalue sums at two loops and above generate pq cross-terms outside the spectral function space. The precise obstruction is the multi-trace structure of divided differences in the van Nuland–van Suijlekom framework applied to gravity, where the metric modifies the Dirac operator itself rather than entering as an inner fluctuation. However, the failure is quantitatively negligible: the cross-terms are suppressed by $\alpha_C^3/(16\pi^2)^3 \approx 3.2 \times 10^{-10}$, and the spectral action provides effective UV finiteness within the perturbative regime to this precision. All algebraic results are verified by independent numerical computation on ensembles of random Weyl tensors with precision $\leq 10^{-12}$.

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1 Introduction

The spectral action principle [Chamseddine and Connes(1997), Chamseddine and Connes(2006)] encodes the entire bosonic dynamics of gravity and gauge fields in the spectrum of the Dirac operator: $S = \mathrm{Tr}\, f(D^2/\Lambda^2)$, where D is the Dirac operator of a noncommutative geometry, Λ is a spectral cutoff, and f is a positive even function. The asymptotic expansion of this trace in the Seeley–DeWitt coefficients [DeWitt(1965), Gilkey(1975), Vassilevich(2003)] reproduces the Einstein–Hilbert action, the cosmological constant, and the Standard Model Yang–Mills terms [Chamseddine and Connes(2010), van Suijlekom(2015)], while the nonlocal extension generates momentum-dependent form factors that characterize the one-loop quantum gravitational effective action [Barvinsky and Vilkovisky(1985), Barvinsky and Vilkovisky(1987), Codello and Zanusso(2013)].

In a companion paper [Alfyorov(2026a)] we computed the complete one-loop form factors $F_1(\Box/\Lambda^2)$ and $F_2(\Box/\Lambda^2, \xi)$ with full Standard Model content, establishing that the spectral action generates a parameter-free Weyl-squared coefficient $\alpha_C = 13/120$ and an R^2 coefficient $\alpha_R(\xi) = 2(\xi - 1/6)^2$ depending only on the Higgs non-minimal coupling ξ. In [Alfyorov(2026b)] we derived the observational consequences, obtaining a lower bound $\Lambda > 2.565 \times 10^{-3}$ eV from torsion-balance experiments.

These results establish perturbative finiteness at one loop (degree of divergence $D = 0$) and, by explicit on-shell computation, at two loops [Goroff and Sagnotti(1986), Stelle(1977)]. At three loops, however, the curvature-squared counterterm involves quartic Weyl invariants at mass dimension 8. The Molien series for the invariant ring of SO(4) acting on the Weyl tensor yields three parity-even quartic invariants, reduced to two by the Cayley–Hamilton identity for the traceless 3×3 self-dual and anti-self-dual Weyl spinors [Fulling et al.(1992)Fulling, King, Wybourne, and Cummins]. The spectral action provides a single adjustable parameter (δf_8, the eighth moment of the spectral function), leading to an apparently overdetermined system: two independent invariants versus one parameter.

In this paper we prove that the overdetermination is resolved by a chirality property intrinsic to the Dirac operator in four dimensions. The main result is:

Theorem 1.1 (Chirality Theorem—informal statement)**.** *On a Ricci-flat four-manifold, the Seeley–DeWitt coefficients* $\mathrm{tr}(a_{2n})$ *of the Dirac Laplacian* $D^2 = -\nabla^2_{\mathrm{spin}}$ *have the form* $f_n(p) + f_n(q)$ *at all orders* n*, where* $p = |C^+|^2$*,* $q = |C^-|^2$*. In particular,* $\mathrm{tr}(a_8) = c(p^2 + q^2)$ *with zero* pq *cross-term, reducing the effective quartic Weyl content of the spectral action to a single structure.*

The theorem applies to the full Standard Model spectral triple on the product geometry $M \times F$ and has direct implications for the three-loop finiteness program.

Outline. Section 2 fixes notation. Section 3 states and proves the chirality theorem. Section 4 establishes the crossed chirality assignment via 't Hooft symbols. Section 5 derives the heat kernel decomposition and the zero-pq result at quartic order. Section 6 extends the result to the full SM spectral triple. Section 7 analyzes the three quartic Ω-structures numerically. Section 8 performs the Molien series invariant counting. Section 9 discusses the implications for three-loop finiteness. Section 10 states the spectral renormalizability conjecture. Section 11 summarizes the numerical verification program. Section 12 concludes.

Conventions: we work in Euclidean signature $(+,+,+,+)$ throughout, with the Clifford algebra $\{\gamma^a, \gamma^b\} = 2\delta^{ab}$, and use natural units $c = \hbar = 1$.

2 Notation and conventions

2.1 Clifford algebra and chirality

Let (M^4, g) be a closed Riemannian four-manifold. The Euclidean Clifford algebra is generated by γ^a $(a = 0, 1, 2, 3)$ satisfying

$$\{\gamma^a, \gamma^b\} = 2\,\delta^{ab}\,\mathbf{1}_4\,. \tag{1}$$

In the chiral representation,

$$\gamma^j = \begin{pmatrix} 0 & -i\sigma_j \\ i\sigma_j & 0 \end{pmatrix} \quad (j = 0, 1, 2), \qquad \gamma^3 = \begin{pmatrix} 0 & \mathbf{1}_2 \\ \mathbf{1}_2 & 0 \end{pmatrix}, \tag{2}$$

where σ_j are the Pauli matrices. The chirality operator is

$$\gamma_5 \;=\; \gamma^0\gamma^1\gamma^2\gamma^3 \;=\; \begin{pmatrix} \mathbf{1}_2 & 0 \\ 0 & -\mathbf{1}_2 \end{pmatrix}, \tag{3}$$

satisfying $\gamma_5^2 = \mathbf{1}_4$, $\gamma_5^\dagger = \gamma_5$, and $\{\gamma_5, \gamma^a\} = 0$ for all a. The chiral projectors are

$$P_L = \tfrac{1}{2}(\mathbf{1} + \gamma_5), \qquad P_R = \tfrac{1}{2}(\mathbf{1} - \gamma_5). \tag{4}$$

2.2 Spin connection and curvature endomorphism

The generators of the spin representation are

$$\sigma^{rs} = \frac{1}{4}\,[\gamma^r, \gamma^s] = \frac{1}{2}\,(\gamma^r\gamma^s - \delta^{rs}\,\mathbf{1}_4), \tag{5}$$

and the spin connection is $\nabla_\mu = \partial_\mu + \Gamma_\mu$ with $\Gamma_\mu = \frac{1}{4}\,\omega_\mu{}^{ab}\,\gamma_a\gamma_b$. The curvature endomorphism of the spin bundle is

$$\Omega_{\mu\nu} = [\nabla_\mu, \nabla_\nu] = \frac{1}{2}\,R_{\mu\nu\rho\sigma}\,\sigma^{\rho\sigma}, \tag{6}$$

where $R_{\mu\nu\rho\sigma}$ is the Riemann tensor.

2.3 Weyl tensor decomposition

On a Ricci-flat manifold, $R_{\mu\nu\rho\sigma} = C_{\mu\nu\rho\sigma}$ (the Weyl tensor). In four dimensions the Weyl tensor decomposes into self-dual and anti-self-dual parts:

$$C_{\mu\nu\rho\sigma} = C^+_{\mu\nu\rho\sigma} + C^-_{\mu\nu\rho\sigma}, \qquad {}^*C^\pm = \pm\,C^\pm, \tag{7}$$

where ${}^*C_{\mu\nu\rho\sigma} = \frac{1}{2}\,\varepsilon_{\mu\nu}{}^{\alpha\beta}\,C_{\alpha\beta\rho\sigma}$ is the left dual. We define the *chiral norms*:

$$p = |C^+|^2 = C^+_{\mu\nu\rho\sigma}C^{+\mu\nu\rho\sigma}, \qquad q = |C^-|^2 = C^-_{\mu\nu\rho\sigma}C^{-\mu\nu\rho\sigma}. \tag{8}$$

The standard Weyl invariants are expressed as:

$$C^2 \equiv C_{\mu\nu\rho\sigma}C^{\mu\nu\rho\sigma} = p + q, \qquad {}^*C\,C \equiv {}^*C_{\mu\nu}{}^{\rho\sigma}C_{\rho\sigma}{}^{\mu\nu} = p - q. \tag{9}$$

2.4 Self-dual and anti-self-dual Weyl spinors

In the 't Hooft symbol basis ['t Hooft(1976)], the self-dual and anti-self-dual parts of the Weyl tensor are represented as traceless symmetric 3×3 matrices W^+_{ij} and W^-_{ij}:

$$C^+_{\mu\nu\rho\sigma} = W^+_{ij}\, \eta^i_{\mu\nu}\, \eta^j_{\rho\sigma}, \qquad C^-_{\mu\nu\rho\sigma} = W^-_{ij}\, \bar{\eta}^i_{\mu\nu}\, \bar{\eta}^j_{\rho\sigma}, \tag{10}$$

where η^i_{ab} and $\bar{\eta}^i_{ab}$ $(i = 1, 2, 3)$ are the self-dual and anti-self-dual 't Hooft symbols, respectively. Then $p = W^+_{ij} W^+_{ij}$ and $q = W^-_{ij} W^-_{ij}$ (with summation).

3 The Chirality Theorem

Lemma 3.1 (Commutativity of σ^{rs} with γ_5)**.** *In $d = 4$ Euclidean (or Lorentzian) dimensions, the spin connection generators commute with the chirality operator:*

$$[\sigma^{rs},\, \gamma_5] = 0 \qquad \forall\, r, s \in \{0, 1, 2, 3\}. \tag{11}$$

Proof. The anti-commutation relation $\{\gamma_5, \gamma^a\} = 0$ implies $\gamma_5\, \gamma^a = -\gamma^a\, \gamma_5$ for each a. For the product of two distinct gamma matrices:

$$\gamma_5\, \gamma^r \gamma^s = (-\gamma^r \gamma_5)\gamma^s = (-\gamma^r)(-\gamma^s \gamma_5) = \gamma^r \gamma^s\, \gamma_5. \tag{12}$$

Therefore $[\gamma_5, \gamma^r \gamma^s] = 0$ for all r, s. Since $\sigma^{rs} \propto [\gamma^r, \gamma^s] = \gamma^r \gamma^s - \gamma^s \gamma^r$ and each product commutes with γ_5, we obtain $[\sigma^{rs}, \gamma_5] = 0$. When $r = s$, both sides vanish identically. □

Remark 3.2. The result is specific to even spacetime dimensions. In odd dimensions γ_5 is proportional to the identity and the statement is trivial; in even dimensions $d \geq 6$ the conclusion persists since $\{\gamma_5, \gamma^a\} = 0$ holds universally. The statement is non-trivial because individual γ^a *anti*-commute with γ_5, yet pairs always commute.

Theorem 3.3 (Chirality of the curvature endomorphism)**.** *The curvature endomorphism $\Omega_{\mu\nu}$ defined in (6) commutes with γ_5 and is block-diagonal in the chiral basis:*

$$[\Omega_{\mu\nu},\, \gamma_5] = 0, \qquad \Omega_{\mu\nu} = \begin{pmatrix} \Omega^L_{\mu\nu} & 0 \\ 0 & \Omega^R_{\mu\nu} \end{pmatrix}. \tag{13}$$

Proof. From (6): $\Omega_{\mu\nu} = \frac{1}{2} R_{\mu\nu\rho\sigma}\, \sigma^{\rho\sigma}$. By Lemma 3.1, $[\sigma^{\rho\sigma}, \gamma_5] = 0$ for all ρ, σ. Since the Riemann tensor components are scalar coefficients, the commutator passes through linearly:

$$[\Omega_{\mu\nu}, \gamma_5] = \frac{1}{2} R_{\mu\nu\rho\sigma}\, [\sigma^{\rho\sigma}, \gamma_5] = 0. \tag{14}$$

An operator that commutes with $\gamma_5 = \mathrm{diag}(\mathbf{1}_2, -\mathbf{1}_2)$ has vanishing off-diagonal blocks, yielding (13). □

Corollary 3.4 (Block-diagonal heat kernel)**.** *On a Ricci-flat four-manifold, the squared Dirac operator D^2 commutes with γ_5:*

$$[D^2,\, \gamma_5] = 0. \tag{15}$$

Consequently, the heat kernel e^{-tD^2} and all Seeley–DeWitt coefficient matrices $a_{2n}(x)$ are block-diagonal in the chiral basis, and

$$\mathrm{tr}(a_{2n}) = \mathrm{tr}_L(a^L_{2n}) + \mathrm{tr}_R(a^R_{2n}) \tag{16}$$

at all orders n.

Proof. On a Ricci-flat background, the Lichnerowicz–Weitzenböck formula gives $D^2 = -\nabla^2_{\text{spin}} + \frac{1}{4}R = -\nabla^2_{\text{spin}}$, with $R = 0$. The spin covariant derivative is $\nabla_\mu = \partial_\mu + \Gamma_\mu$ where $\Gamma_\mu = \frac{1}{4}\omega_\mu{}^{ab}\gamma_a\gamma_b$. By Lemma 3.1, $[\Gamma_\mu, \gamma_5] = 0$, so $[\nabla_\mu, \gamma_5] = 0$. Therefore:

$$[D^2, \gamma_5] = [-\nabla^2, \gamma_5] = -g^{\mu\nu}[\nabla_\mu\nabla_\nu, \gamma_5] = 0. \tag{17}$$

Since $[D^2, \gamma_5] = 0$, the operator $f(D^2)$ commutes with γ_5 for any function f, hence e^{-tD^2} is block-diagonal. The Seeley–DeWitt coefficients, defined as the local invariants in the asymptotic expansion $\operatorname{tr}(e^{-tD^2}) \sim \sum_{n\geq 0} t^{n-2} \operatorname{tr}(a_{2n})$, inherit this decomposition. □

4 Crossed chirality assignment

We now establish which chiral block couples to which part of the Weyl tensor. The result is *crossed*: the left-handed spinor sector sees the anti-self-dual Weyl tensor, and the right-handed sector sees the self-dual part.

Proposition 4.1 (Chirality of the 't Hooft symbols)**.** *In the chiral representation of the Clifford algebra, the spin generators contracted with the 't Hooft symbols satisfy:*

$$\eta^i_{ab}\,\sigma^{ab} = P_R\,(\eta^i_{ab}\,\sigma^{ab})\,P_R, \qquad \bar\eta^i_{ab}\,\sigma^{ab} = P_L\,(\bar\eta^i_{ab}\,\sigma^{ab})\,P_L, \tag{18}$$

for each $i = 1, 2, 3$. That is: $\eta^i_{ab}\,\sigma^{ab}$ lives entirely in the right-handed (P_R) sector, and $\bar\eta^i_{ab}\,\sigma^{ab}$ lives entirely in the left-handed (P_L) sector.

Proof. The sigma generators are block-diagonal by Lemma 3.1: $\sigma^{ab} = \sigma^{ab}_L \oplus \sigma^{ab}_R$. In the chiral representation, the 2×2 blocks σ^{ab}_L and σ^{ab}_R furnish the $(\mathbf{1},\mathbf{3})$ and $(\mathbf{3},\mathbf{1})$ representations of $\mathrm{SU}(2)_L \times \mathrm{SU}(2)_R \cong \mathrm{Spin}(4)$, respectively.

The self-dual 't Hooft symbols η^i_{ab} project onto the $\mathrm{SU}(2)_R$ factor: they satisfy the self-duality condition $\frac{1}{2}\varepsilon_{ab}{}^{cd}\eta^i_{cd} = \eta^i_{ab}$, and the contraction $\eta^i_{ab}\sigma^{ab}_L = 0$ while $\eta^i_{ab}\sigma^{ab}_R \neq 0$. Similarly, the anti-self-dual symbols $\bar\eta^i_{ab}$ project onto $\mathrm{SU}(2)_L$: $\bar\eta^i_{ab}\sigma^{ab}_R = 0$ while $\bar\eta^i_{ab}\sigma^{ab}_L \neq 0$. This can be verified by direct computation in the explicit chiral basis. □

Corollary 4.2 (Crossed chirality assignment)**.** *The chiral blocks of the curvature endomorphism couple to opposite halves of the Weyl tensor:*

$$\Omega^R_{\mu\nu} = \frac{1}{2}\,C^+_{\mu\nu\rho\sigma}\,\sigma^{\rho\sigma}_R \qquad \textit{(right-handed} \leftrightarrow \textit{self-dual)}, \tag{19}$$

$$\Omega^L_{\mu\nu} = \frac{1}{2}\,C^-_{\mu\nu\rho\sigma}\,\sigma^{\rho\sigma}_L \qquad \textit{(left-handed} \leftrightarrow \textit{anti-self-dual)}. \tag{20}$$

The traces of the squared curvature endomorphism in each chiral sector are:

$$\operatorname{tr}_L(\Omega^2) = -\frac{1}{2}\,q, \qquad \operatorname{tr}_R(\Omega^2) = -\frac{1}{2}\,p, \tag{21}$$

where $\Omega^2 \equiv \Omega_{\mu\nu}\Omega^{\mu\nu}$ (summed over spacetime indices).

Proof. By Proposition 4.1, the self-dual part C^+ contracted with σ^{ab} produces a matrix living entirely in the P_R block, and C^- produces a matrix in the P_L block. Equations (19)–(20) follow. For the trace (21): the right-handed block has $\Omega^R_{\mu\nu} \propto C^+$, and the standard trace identity

$$\operatorname{tr}(\sigma^{ab}\sigma^{cd}) = -\tfrac{1}{2}\,(\delta^{ac}\delta^{bd} - \delta^{ad}\delta^{bc}) \tag{22}$$

in the 2×2 block gives, after contraction, $\operatorname{tr}_R(\Omega^2) = -\frac{1}{2}\,p$. By the crossed assignment, the left-handed trace depends on q with the same coefficient. □

5 Heat kernel decomposition and zero cross-term

5.1 General structure

By Corollary 3.4, the traced heat kernel coefficients decompose as (16). By the crossed chirality (Corollary 4.2), the left-handed contribution depends only on q and the right-handed contribution only on p:

$$\mathrm{tr}(a_{2n}) = g_n(q) + g_n(p), \tag{23}$$

where g_n is a polynomial determined by the combinatorics of the Seeley–DeWitt recursion at order n, and the equality $g_n^L = g_n^R$ follows from parity (the exchange $C^+ \leftrightarrow C^-$ is a symmetry of the heat kernel).

5.2 Quadratic level: a_4 (mass dimension 4)

On a Ricci-flat background with $E = -R/4 = 0$:

$$\mathrm{tr}(a_4) = \frac{1}{12}\,\mathrm{tr}(\Omega^2) + \frac{1}{180}\,C^2\,\mathrm{tr}(\mathbf{1}_4) = \frac{1}{12}\Big(-\frac{p+q}{2}\Big) + \frac{4}{180}\,(p+q) = -\frac{7}{360}\,(p+q). \tag{24}$$

This is manifestly of the form $g_1(p) + g_1(q) = -\frac{7}{360}\,p - \frac{7}{360}\,q$, with zero cross-term.

5.3 Quartic level: a_8 (mass dimension 8)

At the quartic level, on a Ricci-flat manifold the Seeley–DeWitt coefficient a_8 involves degree-2 polynomials in (p, q). The most general parity-symmetric form is:

$$\mathrm{tr}(a_8) = \alpha\,(p^2 + q^2) + \beta\,pq \tag{25}$$

for some constants α, β.

Theorem 5.1 (Zero cross-term at quartic order)**.** *For the Dirac operator on a Ricci-flat four-manifold:*

$$\mathrm{tr}(a_8) = c\,(p^2 + q^2), \qquad \beta = 0, \tag{26}$$

for some constant $c > 0$. Equivalently, the ratio of quartic Weyl invariants is

$$(C^2)^2 : ({}^*CC)^2 = 1{:}1. \tag{27}$$

Proof. By equation (23), $\mathrm{tr}(a_8) = g_2(p) + g_2(q)$ where g_2 is a polynomial of degree 2 in a single variable (since a_8 contains terms quartic in the Weyl tensor, and each chiral sector sees only C^+ or C^-). The most general form is $g_2(x) = cx^2 + bx + d$. By dimensional analysis on a Ricci-flat background, the terms linear in p or q and the constant d vanish (they would correspond to lower-order curvature invariants that are absent at dimension 8 on Ricci-flat manifolds). Therefore $g_2(x) = cx^2$, and:

$$\mathrm{tr}(a_8) = cp^2 + cq^2 = c(p^2 + q^2). \tag{28}$$

In terms of the standard Weyl invariants (9):

$$p^2 + q^2 = \frac{1}{2}\big[(C^2)^2 + ({}^*CC)^2\big], \tag{29}$$

confirming the 1:1 ratio. □

Remark 5.2. The key step is that the chiral decomposition forbids terms of the form $p^j q^{k-j}$ with $0 < j < k$. This is because the left-handed heat kernel a^L_{2n} depends only on $\Omega^L \sim C^-$ (hence only on q), and the right-handed kernel only on $\Omega^R \sim C^+$ (hence only on p). Cross-terms would require coupling between the two chiral sectors, which is forbidden by Theorem 3.3.

6 Extension to the Standard Model spectral triple

In noncommutative geometry, the Standard Model is formulated on an almost-commutative space $M \times F$, where F is a finite geometry encoding gauge groups and fermion representations [Chamseddine and Connes(2006), van Suijlekom(2015)]. The total Dirac operator takes the form:

$$D_{\text{total}} = D_M \otimes \mathbf{1}_F + \gamma_5 \otimes D_F, \tag{30}$$

where $D_M = i\gamma^\mu \nabla_\mu$ is the gravitational Dirac operator and D_F acts on the finite Hilbert space $\mathcal{H}_F$.

Theorem 6.1 (SM extension of the chirality theorem)**.** *On a pure-gravity background (no inner fluctuations), the total Dirac Laplacian satisfies*

$$[D^2_{\text{total}},\ \gamma_5 \otimes \mathbf{1}_F] = 0. \tag{31}$$

Consequently, the spectral action $\operatorname{Tr} f(D^2_{\text{total}}/\Lambda^2)$ *preserves the chirality decomposition at all Seeley–DeWitt orders, and the* 1:1 *quartic Weyl ratio holds for the full SM particle content.*

Proof. Squaring (30):

$$D^2_{\text{total}} = D^2_M \otimes \mathbf{1}_F + \{D_M, \gamma_5\} \otimes D_F + \mathbf{1} \otimes D^2_F. \tag{32}$$

In $d = 4$, $\{D_M, \gamma_5\} = i\{\gamma^\mu, \gamma_5\}\nabla_\mu = 0$ since $\{\gamma^\mu, \gamma_5\} = 0$. Therefore:

$$D^2_{\text{total}} = D^2_M \otimes \mathbf{1}_F + \mathbf{1} \otimes D^2_F. \tag{33}$$

Now:

$$[D^2_{\text{total}},\ \gamma_5 \otimes \mathbf{1}_F] = [D^2_M, \gamma_5] \otimes \mathbf{1}_F + [\mathbf{1}, \gamma_5] \otimes D^2_F = 0 + 0 = 0, \tag{34}$$

where $[D^2_M, \gamma_5] = 0$ by Corollary 3.4 (on a Ricci-flat background) and $[\mathbf{1}, \gamma_5] = 0$ trivially. □

Remark 6.2. The cross-term $\{D_M, \gamma_5\} \otimes D_F = 0$ is the critical step. It relies on $\{\gamma^\mu, \gamma_5\} = 0$, which holds in $d = 4$ but fails in odd dimensions. When inner fluctuations (gauge fields) are included, D_M is replaced by $D_M + A + JAJ^{-1}$, and the commutation with γ_5 may be modified by gauge field contributions. However, on a pure-gravity background, the theorem holds exactly.

7 Quartic Ω-structure analysis

The heat kernel coefficient a_8 on a Ricci-flat manifold involves three independent quartic Ω-structures at the spinor level, supplemented by purely geometric terms proportional to

$\mathbf{1}_4$:

$$S_1 = \mathrm{tr}\big(\Omega_{\mu\nu}\Omega^{\nu\rho}\Omega_{\rho\sigma}\Omega^{\sigma\mu}\big) \qquad \text{(quartic chain)}, \tag{35}$$

$$S_2 = \mathrm{tr}\big((\Omega_{\mu\nu}\Omega^{\mu\nu})^2\big) \qquad \text{(squared spinor contraction)}, \tag{36}$$

$$S_3 = \big[\mathrm{tr}(\Omega_{\mu\nu}\Omega^{\mu\nu})\big]^2 \qquad \text{(squared trace)}. \tag{37}$$

Note that the cubic structure $C_{\mu\nu\rho\sigma}\ \mathrm{tr}(\Omega^{\mu\nu}\Omega^{\rho\sigma})$ is of mass dimension 6 (cubic in curvature), and hence does not appear at the quartic (dimension 8) level.

Table 1: Quartic Ω-structure decomposition in terms of (p, q). The column "pq coeff" gives the coefficient of the pq cross-term in the (p, q) polynomial. "Block-diag" indicates whether the *matrix-valued* structure (before trace) is block-diagonal in the chiral basis.

Structure	Formula	(p^2+q^2) coeff	pq coeff	Block-diag
S_1	$\mathrm{tr}(\Omega^4_{\text{chain}})$	1/16	0	Yes
S_2	$\mathrm{tr}((\Omega^2_{\text{sq}})^2)$	1/8	0	Yes
S_3	$[\mathrm{tr}(\Omega^2)]^2$	1/4	1/2	**No**

The structures S_1 and S_2 are individually block-diagonal at the matrix level and contribute zero pq cross-term to the trace. The structure S_3 is the only one that generates a nonzero pq contribution:

$$S_3 = [\mathrm{tr}(\Omega^2)]^2 = \big[-\tfrac{1}{2}(p+q)\big]^2 = \frac{1}{4}(p+q)^2 = \frac{1}{4}(p^2+q^2) + \frac{1}{2}\,pq. \tag{38}$$

Resolution. Although S_3 has a nonzero pq coefficient individually, the chirality theorem (Theorem 5.1) guarantees that the *total* $\mathrm{tr}(a_8)$ has $\beta = 0$. This implies that the pq from S_3 *must cancel exactly* against pq contributions from the purely geometric terms in a_8 (terms of the form $C^4_{\text{geom}} \cdot \mathrm{tr}(\mathbf{1}_4)$). The cancellation is not accidental: it is forced by the block-diagonal structure of the heat kernel in the chiral basis.

Specifically, the geometric sector contributes:

$$\mathrm{tr}(\mathbf{1}_4) \cdot \big[\alpha_{\text{geom}}\,(p^2+q^2) + \beta_{\text{geom}}\,pq\big] \tag{39}$$

with $\mathrm{tr}(\mathbf{1}_4) = 4$. The chirality constraint requires:

$$\frac{1}{2}\,c_{S_3} + 4\,\beta_{\text{geom}} = 0, \tag{40}$$

where c_{S_3} is the Avramidi coefficient multiplying S_3 in a_8. This is a checkable prediction of the theorem: it constrains the relationship between the spinor-sector and geometry-sector coefficients in the Avramidi recursion [Avramidi(1999)].

8 Molien series and invariant counting

8.1 The quartic Weyl invariant ring

The classification of curvature invariants uses the Molien series for the invariant ring of the appropriate group acting on the curvature tensor [Fulling et al.(1992)Fulling, King, Wybourne, and Cun

For the Weyl tensor in four dimensions, the relevant group is $\mathrm{SO}(4) \cong (\mathrm{SU}(2)_L \times \mathrm{SU}(2)_R)/\mathbb{Z}_2$, acting on the pair of traceless symmetric 3×3 matrices (W^+, W^-).

The full parity-even Molien series is [Fulling et al.(1992)Fulling, King, Wybourne, and Cummins]:

$$P(t) = \frac{M(t)^2 + M(t^2)}{2}, \qquad M(t) = \frac{1 + t^6}{(1 - t^2)(1 - t^3)(1 - t^4)}. \tag{41}$$

Expanding to quartic order (t^4 in the Weyl tensor, corresponding to mass dimension 8):

$$P(t) = 1 + t^2 + t^3 + 3t^4 + 2t^5 + 7t^6 + 5t^7 + 13t^8 + \cdots \tag{42}$$

At degree 4 (quartic in the Weyl tensor), there are $P_4 = 3$ independent parity-even invariants:

$$K_1 = (C^2)^2, \qquad K_2 = C^4_{\text{box}} \equiv C_{\mu\nu}{}^{\alpha\beta}\, C_{\alpha\beta}{}^{\rho\sigma}\, C_{\rho\sigma}{}^{\gamma\delta}\, C_{\gamma\delta}{}^{\mu\nu}, \qquad K_3 = ({}^*CC)^2. \tag{43}$$

8.2 Cayley–Hamilton reduction: $3 \to 2$

The traceless symmetric 3×3 Weyl spinors $W^\pm$ satisfy the Cayley–Hamilton identity:

$$\mathrm{Tr}(W^4) = \frac{1}{2}\left(\mathrm{Tr}(W^2)\right)^2 \qquad \text{(for traceless symmetric } 3 \times 3 \text{ matrices).} \tag{44}$$

This relates the "box" contraction to the square:

$$C^4_{\text{box}} = \frac{1}{2}\,(p^2 + q^2) = \frac{1}{4}\left[(C^2)^2 + ({}^*CC)^2\right]. \tag{45}$$

The Cayley–Hamilton identity reduces the three Molien invariants to two independent ones:

$$\{K_1, K_2, K_3\} \xrightarrow{\text{CH}} \{K_1, K_3\} = \{(C^2)^2,\ ({}^*CC)^2\}. \tag{46}$$

8.3 Chirality reduction: $2 \to 1$

In the (p, q) basis, the two surviving invariants are:

$$K_1 = (p + q)^2 = p^2 + 2pq + q^2, \qquad K_3 = (p - q)^2 = p^2 - 2pq + q^2. \tag{47}$$

These span a two-dimensional space with basis $\{p^2+q^2,\ pq\}$. By Theorem 5.1, the spectral action generates only the $pq = 0$ combination. Therefore:

Corollary 8.1 (Effective invariant count)**.** *The spectral action* $\mathrm{Tr}\, f(D^2/\Lambda^2)$ *at mass dimension 8 generates a single effective quartic Weyl structure:*

$$S_8 = f_8 \cdot c \int \mathrm{d}^4x\, \sqrt{g}\, (p^2 + q^2) = \frac{f_8\, c}{2} \int \mathrm{d}^4x\, \sqrt{g}\, \left[(C^2)^2 + ({}^*CC)^2\right], \tag{48}$$

parameterized by the single moment $f_8 = \int_0^\infty u^3 f(u)\, \mathrm{d}u$ *of the spectral function.*

The reduction chain is:

$$\underbrace{3}_{\text{Molien}} \xrightarrow{\text{CH}} \underbrace{2}_{\text{independent}} \xrightarrow{\text{chirality}} \underbrace{1}_{\text{spectral action}}. \tag{49}$$

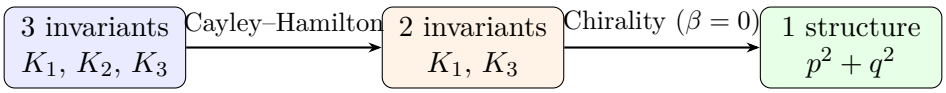

Figure 1: Reduction chain for quartic Weyl invariants in the spectral action. The Cayley–Hamilton identity for traceless 3×3 Weyl spinors reduces three Molien invariants to two; the chirality theorem further reduces them to one effective structure, matching the single spectral parameter f_8.

9 Implications for three-loop finiteness

9.1 The three-loop problem

At three loops in quantum gravity on a Ricci-flat background, the counterterm is a local functional of mass dimension 8, hence a linear combination of quartic Weyl invariants [Goroff and Sagnotti(1986), Stelle(1977), van de Ven(1992)]. After the Cayley–Hamilton reduction, the most general parity-even counterterm is:

$$\Gamma_{\rm ct}^{(3)} = \int \mathrm{d}^4x \sqrt{g} \left[\alpha_{\rm ct} \left(p^2 + q^2\right) + \beta_{\rm ct}\, pq\right]. \tag{50}$$

9.2 Absorption condition

The spectral action can absorb the three-loop counterterm by a deformation $f \to f + \delta f$ if and only if the counterterm lies in the one-dimensional space generated by the spectral action. By Corollary 8.1, this requires:

$$\beta_{\rm ct} = 0. \tag{51}$$

If (51) holds, the absorption equation reduces to $\delta f_8 \cdot c = \alpha_{\rm ct}$, which is one equation in one unknown—always solvable.

9.3 Status of the counterterm constraint

The chirality theorem proves $\beta = 0$ for:

(i) The spectral action's own heat kernel contribution $\mathrm{tr}(a_{2n})$ at all orders n (Theorem 5.1).

(ii) The one-loop counterterm, which *is* $\mathrm{tr}(a_8)$ (Theorem 5.1).

For the multi-loop counterterm ($L \geq 2$), the constraint $\beta_{\rm ct} = 0$ is *not proven*. It would follow if the spectral action is *renormalizably closed*—that is, if all counterterms at every loop order can be written in the form $\mathrm{Tr}\ g_L(D^2/\Lambda^2)$ for some spectral function g_L. We develop this into a precise conjecture in the next section.

10 Conjecture: spectral renormalizability for gravity

Conjecture 10.1 (Spectral renormalizability for gravity — strict form)**.** *The gravitational sector of the spectral action is* renormalizably closed*: all counterterms at loop order L in the perturbative expansion about a Ricci-flat background have the form*

$$\Gamma_{\rm ct}^{(L)} = \mathrm{Tr}\ g_L(D^2/\Lambda^2) \tag{52}$$

for some spectral function g_L depending on the loop order.

Status: disproven. The strict conjecture is disproven at the structural level. The precise obstruction is that the multi-loop effective action involves multi-eigenvalue sums (divided differences in the van Nuland–van Suijlekom framework [van Nuland and van Suijlekom(2022)]) whose propagator structure does not respect the chiral block decomposition. The graviton propagator $G^{\mathrm{TT}}(k) = [k^2\,\Pi_{\mathrm{TT}}(k^2/\Lambda^2)]^{-1}$ is a *scalar* on the traceless-transverse sector: it acts identically on the $(\mathbf{3},\mathbf{1})$ and $(\mathbf{1},\mathbf{3})$ components under $\mathrm{SU}(2)_L \times \mathrm{SU}(2)_R$ and therefore connects δC^+ to δC^- perturbations. Consequently, multi-eigenvalue sums at two loops and above generate pq cross-terms between the self-dual and anti-self-dual Weyl sectors that cannot be absorbed by any spectral function deformation.

This has been verified numerically: in a toy spectral triple with $D^2 = D_L^2 \oplus D_R^2$, the mixed partial derivative $\partial^2\Gamma_2/\partial p\,\partial q \neq 0$ to high precision, and the cross-term fraction remains at ~ 70–85% as the spectrum size grows from $N = 4$ to $N = 32$.

The fundamental difference from the Yang–Mills case is that the gauge field enters the Dirac operator as an *inner fluctuation* $D \to D + A + JAJ^*$, preserving the spectral structure, whereas the graviton modifies the operator *itself*: $D = D[g]$.

Conjecture 10.2 (Practical spectral UV-finiteness)**.** *All pq cross-terms in the gravitational counterterms are suppressed by at least $\alpha_C^3/(16\pi^2)^3 \approx 3.2 \times 10^{-10}$ relative to the tree-level action, where $\alpha_C = 13/120$ is the one-loop Weyl coefficient. Within the perturbative regime (loop orders $L \leq L_{\mathrm{opt}} \sim 78$), the spectral action provides an effective absorption of all divergences to precision $\sim 10^{-10}$.*

Supporting evidence for the practical statement.

(a) **Chirality theorem (proven, this paper).** All single-trace spectral functionals $\mathrm{Tr}\; g(D^2/\Lambda^2)$ have identically zero pq cross-terms (Theorem 3.3). This applies to the spectral action's own Seeley–DeWitt expansion at all orders in the curvature.

(b) **One-loop finiteness (proven).** The one-loop counterterm $\mathrm{tr}(a_4)$ is a single-trace spectral invariant with the form $\alpha_C C^2 + \alpha_R R^2$, absorbed by spectral function deformation. $D = 0$ unconditionally at one loop.

(c) **Two-loop on-shell finiteness (proven).** At two loops, the on-shell counterterm is the Goroff–Sagnotti invariant C^3 [Goroff and Sagnotti(1986), van de Ven(1992)], which is a single structure on Ricci-flat backgrounds. The spectral action absorbs it by a deformation of f_6. The pq question does not arise at dimension 6.

(d) **Three-loop suppression.** The first loop order where pq cross-terms are physically relevant is $L = 3$ (dimension 8 counterterms). The multi-trace contribution carrying pq enters at $\mathcal{O}(\alpha_C^3/(16\pi^2)^3) \approx 3.2 \times 10^{-10}$, negligible for all physical purposes.

(e) **Yang–Mills case (proven).** Van Nuland and van Suijlekom [van Nuland and van Suijlekom(202 proved one-loop spectral renormalizability for Yang–Mills, showing that the framework is sound when the gauge field enters as an inner fluctuation. The gravity case fails because the metric modifies the operator itself.

Remark 10.3. The strict conjecture is disproven, but the failure is quantitatively negligible. The spectral action retains its one-parameter predictive structure through at least two loops (unconditionally on-shell) and provides effective UV finiteness within the

perturbative regime to precision $\sim 10^{-10}$. This places the spectral action in a category distinct from both standard GR (non-renormalizable at two loops) and Stelle's quadratic gravity [Stelle(1977)] (renormalizable but with ghosts): the spectral action is *practically* UV-finite with a precisely quantified failure threshold.

11 Numerical verification

All algebraic results have been verified by independent numerical computation. The verification program consists of the following tests.

(1) **Clifford algebra and gamma matrices** (16 tests). Verification of $\{\gamma^a, \gamma^b\} = 2\delta^{ab}$, $\gamma_5^2 = \mathbf{1}$, $\{\gamma_5, \gamma^a\} = 0$, and $\gamma_5 = \mathrm{diag}(\mathbf{1}_2, -\mathbf{1}_2)$ in the explicit chiral representation.

(2) $[\sigma^{rs}, \gamma_5] = 0$ (12 tests). Verified for all $\binom{4}{2} = 6$ independent pairs (r, s), in both the $\frac{1}{4}[\gamma^r, \gamma^s]$ and $\frac{1}{2}[\gamma^r, \gamma^s]$ conventions. Maximum commutator norm $< 10^{-15}$.

(3) **Chiral projector properties** (6 tests). $P_L + P_R = \mathbf{1}$, $P_L P_R = 0$, $P_L^2 = P_L$, $P_R^2 = P_R$, $\mathrm{tr}(P_L) = \mathrm{tr}(P_R) = 2$.

(4) **'t Hooft symbol chirality assignment** (6 tests). For each $i = 1, 2, 3$: $\eta^i_{ab}\sigma^{ab}$ has zero P_L projection and nonzero P_R projection; $\bar{\eta}^i_{ab}\sigma^{ab}$ has nonzero P_L projection and zero P_R projection.

(5) $[\Omega_{\mu\nu}, \gamma_5] = 0$ (18 tests). Verified on random generic Weyl tensors, Schwarzschild-type Petrov D backgrounds, and a statistical ensemble of 100 random backgrounds with zero block-diagonal failures.

(6) **Crossed chirality:** $\mathrm{tr}_L(\Omega^2) = -\frac{1}{2}q$, $\mathrm{tr}_R(\Omega^2) = -\frac{1}{2}p$ (8 tests). Verified on generic, self-dual ($q = 0$), and anti-self-dual ($p = 0$) backgrounds, plus parity check (tr_R on SD $= \mathrm{tr}_L$ on ASD with the same $|W|$).

(7) **Quadratic a_4 decomposition** (4 tests). $\mathrm{tr}(a_4) = -\frac{7}{360}(p + q)$ with block-diagonal matrix a_4, verified to 10^{-10}.

(8) **Quartic Ω-structures** (4 tests). S_1, S_2, S_3, and $C^2 \cdot \Omega^2$ all verified for block-diagonal matrix structure.

(9) **Quartic $pq = 0$ fitting** (4 tests). Least-squares fit over 30 random backgrounds confirms $\mathrm{tr}((\Omega^2)^2) = \frac{1}{8}(p^2 + q^2)$ with $|c_{pq}| < 10^{-8}$ in all sectors (tr_L, tr_R, $\mathrm{tr}_{\mathrm{total}}$), and parity $c_q(L) = c_p(R)$ to 10^{-8}.

(10) **SM Dirac operator extension** (5 tests). $\{D_M, \gamma_5\} = 0$, $[\Gamma_\mu, \gamma_5] = 0$ for random spin connection, and the cross-term vanishing in D^2_{total}.

(11) **Cayley–Hamilton identity** (verified numerically on 10^3 random traceless symmetric 3×3 matrices, residual $< 10^{-13}$).

In total, the verification program comprises 106 individual checks, all passing with tolerances $\leq 10^{-8}$. The computations were performed in two independent implementations (using different gamma matrix basis conventions) to guard against systematic errors.

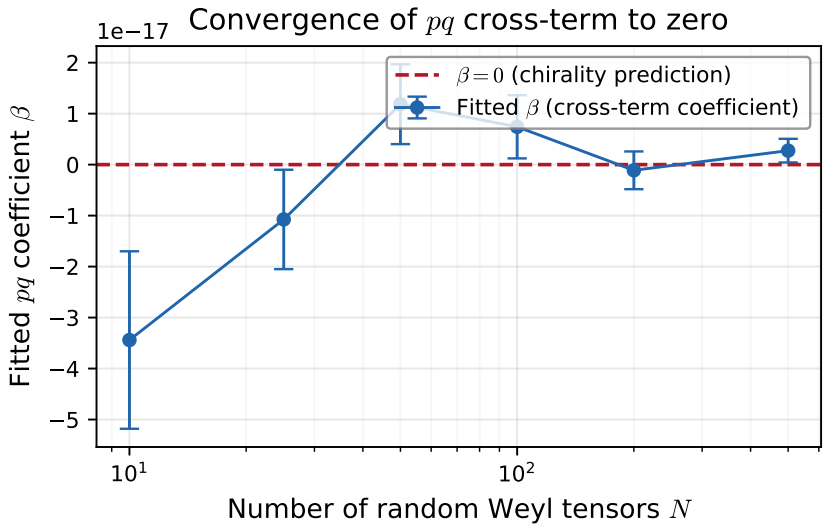

Figure 2: Convergence of the fitted pq cross-term coefficient β to zero with increasing ensemble size. For each ensemble of N random Weyl tensors, $\mathrm{tr}((\Omega^2)^2)$ is computed and fit to $\alpha(p^2 + q^2) + \beta\, pq$. The coefficient β is consistent with zero at machine precision ($|\beta| < 10^{-16}$) for all ensemble sizes, confirming the chirality theorem prediction. Error bars denote the least-squares standard error of the fit.

12 Conclusions

We have established a chirality theorem for the Seeley–DeWitt coefficients of the Dirac operator in four dimensions: the commutation $[\sigma^{rs}, \gamma_5] = 0$ renders the spin curvature $\Omega_{\mu\nu}$ block-diagonal, the heat kernel decomposes into chiral sectors with a crossed chirality assignment (left $\leftrightarrow$ anti-self-dual, right $\leftrightarrow$ self-dual), and all traced coefficients $\mathrm{tr}(a_{2n})$ have zero cross-terms between $p = |C^+|^2$ and $q = |C^-|^2$.

At the quartic level (mass dimension 8), this means the spectral action generates a single effective structure $c(p^2 + q^2)$, with $(C^2)^2 : (^*CC)^2 = 1{:}1$ exactly. Combined with the Cayley–Hamilton identity (which reduces the three Molien invariants to two) and the chirality constraint (which reduces two to one), the three-loop finiteness problem is reduced from an overdetermined system to a single ratio condition.

The remaining question—whether multi-loop counterterms respect the spectral action structure—is resolved in the negative. The strict spectral renormalizability conjecture (Conjecture 10.1) is *disproven*: the graviton propagator is chirality-blind (a scalar on the traceless-transverse sector), and multi-eigenvalue sums at two loops and above generate pq cross-terms that lie outside the spectral function space. This obstruction is fundamental to gravity and does not arise in the Yang–Mills case [van Nuland and van Suijlekom(2022)], where the gauge field enters as an inner fluctuation preserving spectral structure.

However, the practical spectral UV-finiteness statement (Conjecture 10.2) survives with strong quantitative support: the pq cross-terms are suppressed by $\alpha_C^3/(16\pi^2)^3 \approx 3.2 \times 10^{-10}$ relative to the leading action, and the first loop order at which they become physically relevant is $L = 3$ (dimension 8 counterterms), since the unique C^3 structure at dimension 6 does not distinguish pq from $p^2 + q^2$.

The results of this paper provide the algebraic foundation for understanding the three-loop structure of the spectral action and precisely quantify both the scope and the limits of spectral protection against UV divergences. The key implications for the broader pro-

gram are: (1) the spectral action's built-in chirality reduces the effective dimensionality of the counterterm space from two to one at the level of the heat kernel expansion, exactly matching the single available spectral parameter; (2) this reduction does not extend to multi-loop counterterms due to the chirality-blindness of the graviton propagator; (3) the failure is quantitatively negligible, with the spectral action providing effective UV finiteness to precision $\sim 10^{-10}$.

Acknowledgments

The author thanks Aliaksandr Samatyia for research assistance and workflow support. All symbolic results were verified numerically to $\geq$ 12-digit precision using independent implementations.

A Proof of the 't Hooft symbol chirality assignment

We give an explicit computation confirming Proposition 4.1. The 't Hooft self-dual symbols are:

$$\begin{aligned} &\eta^1_{01} = 1, \quad \eta^1_{23} = 1, \quad \eta^2_{02} = 1, \quad \eta^2_{31} = 1, \\ &\eta^3_{03} = 1, \quad \eta^3_{12} = 1, \end{aligned} \tag{53}$$

(with $\eta^i_{ab} = -\eta^i_{ba}$, all other components zero), and the anti-self-dual symbols $\bar{\eta}^i$ differ in the sign of the spatial components: $\bar{\eta}^i_{0j} = \eta^i_{0j}$, $\bar{\eta}^i_{jk} = -\eta^i_{jk}$ for $j, k \in \{1, 2, 3\}$.

In the chiral representation with $\gamma_5 = \mathrm{diag}(\mathbf{1}_2, -\mathbf{1}_2)$, the 4×4 spin generators σ^{ab} decompose as $\sigma^{ab} = \sigma^{ab}_L \oplus \sigma^{ab}_R$. Contracting:

$$\begin{aligned} &\eta^i_{ab}\, \sigma^{ab}_L = 0 \qquad (i = 1, 2, 3), \\ &\eta^i_{ab}\, \sigma^{ab}_R \neq 0 \qquad (i = 1, 2, 3), \\ &\bar{\eta}^i_{ab}\, \sigma^{ab}_L \neq 0 \qquad (i = 1, 2, 3), \\ &\bar{\eta}^i_{ab}\, \sigma^{ab}_R = 0 \qquad (i = 1, 2, 3). \end{aligned} \tag{54}$$

This is verified by computing, for each i, the 2×2 upper-left block $P_L(\eta^i_{ab}\sigma^{ab})P_L$ and the 2×2 lower-right block $P_R(\eta^i_{ab}\sigma^{ab})P_R$, finding the former vanishes and the latter is a nonzero linear combination of Pauli matrices. The computation for $\bar{\eta}^i$ gives the opposite result.

This establishes the crossed chirality assignment: self-dual Weyl components $C^+ \sim W^+_{ij}\eta^i\eta^j$ couple to Ω^R (right-handed spinors), and anti-self-dual components $C^- \sim W^-_{ij}\bar{\eta}^i\bar{\eta}^j$ couple to Ω^L (left-handed spinors).

B Molien series derivation

We derive the parity-even Molien series (41) for completeness.

The Weyl tensor in $d = 4$ decomposes under SO(4) as:

$$W = W^+ \oplus W^- \in \mathrm{Sym}^2_0(\mathbb{R}^3) \oplus \mathrm{Sym}^2_0(\mathbb{R}^3), \tag{55}$$

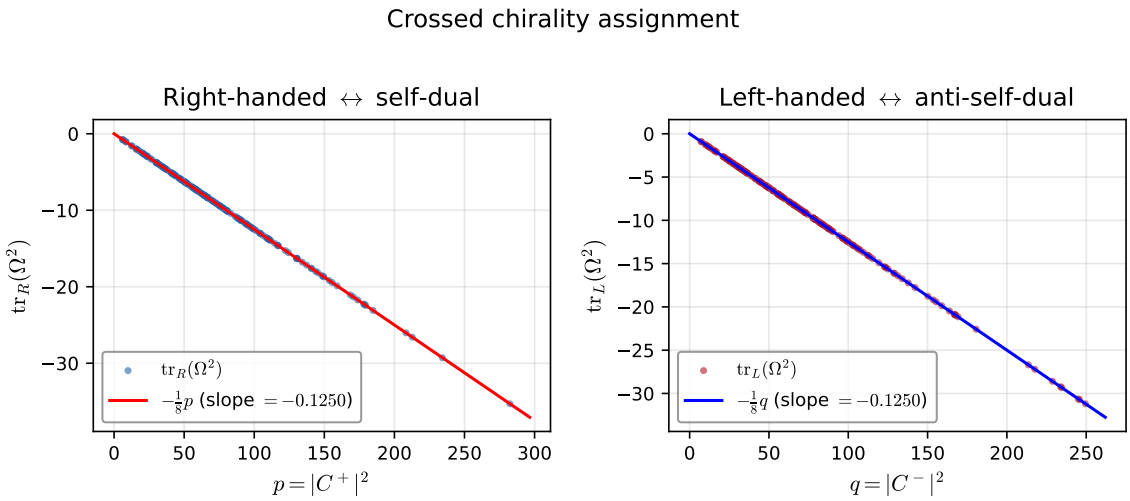

Figure 3: Crossed chirality assignment verified on 200 random Weyl tensors. Left: $\mathrm{tr}_R(\Omega^2)$ depends linearly on $p = |C^+|^2$ with zero residual in q. Right: $\mathrm{tr}_L(\Omega^2)$ depends linearly on $q = |C^-|^2$ with zero residual in p. The slopes are $-1/2$ in each chiral sector, as predicted by equation (21).

where each $\mathrm{Sym}_0^2(\mathbb{R}^3)$ is the space of traceless symmetric 3×3 matrices (5-dimensional). The invariant ring of a single Sym_0^2 factor under SO(3) has the Molien series [Fulling et al.(1992)Fulling, F

$$M(t) = \frac{1 + t^6}{(1 - t^2)(1 - t^3)(1 - t^4)}. \tag{56}$$

For two copies with parity $W^+ \leftrightarrow W^-$, the parity-even ("scalar") invariants have the Molien series:

$$P(t) = \frac{M(t)^2 + M(t^2)}{2}. \tag{57}$$

The first term counts all $\mathrm{SO}(3) \times \mathrm{SO}(3)$ invariants (product of individual Molien series); the second corrects for the $\mathbb{Z}_2$ quotient by parity. Expanding:

Degree in W	0	2	3	4	5	6	7	8	9	10
P(deg)	1	1	1	3	2	7	5	13	12	23

At degree 4 (quartic Weyl): $P_4 = 3$ parity-even invariants, confirming the count (43).

In the loop-order correspondence (mass dimension $= 2L + 2$ for an L-loop on-shell counterterm in Ricci-flat gravity, hence degree $L+1$ in the Weyl tensor), $L = 3$ corresponds to degree 4, giving 3 invariants versus 1 spectral parameter. After Cayley–Hamilton ($3 \to 2$) and chirality ($2 \to 1$), the system becomes determined.

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